

MLLM-Geo-AI

Remaining Phases & Tasks Report

Gap Analysis vs Step 2 Plan · May 4, 2026

Defense Readiness Score	Required Threshold
22 / 50 — NOT READY	≥ 35 / 50

Dimension	Score	Key Problem
ML Training Quality	3/10	Single-class dataset, trivial 100% accuracy
Output Schema Completeness	7/10	2 fields missing from PRD v3.0
API Coverage	6/10	7 endpoints missing
Step 2 Phase Completion	5/10	8/12 phases partial or missing
Demo Stability	1/10	MLP weight mismatch + POI encoder crash

1. Current Status — What Was Fixed vs What Remains

Issue	Previous	Now	Action Required
Class imbalance (all label=0)	BROKEN	BROKEN	Re-label CSV: map category to 0/1/2
Val split unused in training	BROKEN	BROKEN	Add val_loss loop to train_multimodal.py
Graph features hardcoded zeros	BROKEN	FIXED	Done — real OSMnx features
No ablation support (--modalities)	BROKEN	BROKEN	Add argparse flag to training script
Spatial Accuracy metric missing	MISSING	MISSING	Implement 8-neighbor algorithm in eval
MLLM / LLM not integrated	MISSING	MISSING	Planned P2 — mention in defense
Digital Twin NL query (501)	MISSING	MISSING	Planned P2 — mention in defense

2. P0 — Critical Fixes (Must Complete Before Defense)

These 5 issues are hard blockers. The demo will crash or produce meaningless results without them.
Estimated total: ~4 hours.

P0-1 Fix Dataset Labels — Single Class Problem

File: data/raw/project.csv

All 973 rows have label=0 (Residential). The model learns nothing and reports trivial 100% accuracy. You must create real multi-class ground truth.

Fix: remap the category column

```
# Python script - run once to fix the CSV
import pandas as pd
df = pd.read_csv('data/raw/project.csv')

label_map = {
    'Education': 0,    # Residential
    'Health':      0,    # Residential
    # Add more categories from your data:
    # 'mall':      1,    'shop': 1,    'bank': 1,    # Commercial
    # 'factory':   2,    'industrial': 2,          # Industrial
}
df['label'] = df['category'].str.lower().map(label_map).fillna(0).astype(int)
df.to_csv('data/raw/project.csv', index=False)
```

After fixing: verify `df['label'].value_counts()` shows at least 2 classes. You need Commercial (label=1) and Industrial (label=2) POIs in your data. If none exist, create ~100 synthetic rows for each class from realistic POI names.

Time estimate: 2 hours

P0-2 Fix MLP Weight Dimension Mismatch

File: app/domain/mlp_model.py, line 12

Saved model weights use `hidden_dim=256` but the code initializes with `hidden_dim=128`. This causes a shape error at load time.

Fix: change one line

```
# BEFORE (line 12)
def __init__(self, input_dim, hidden_dim=128, output_dim=3):

# AFTER
def __init__(self, input_dim, hidden_dim=256, output_dim=3):
```

OR: delete models/urban_mlp.pt and retrain with `hidden_dim=128` after fixing P0-1.

Time estimate: 30 minutes

P0-3 Fix POI Encoder API Error (encode vs embed_texts)

Files: scripts/train_multimodal.py line 27, and any other callers

The code calls `poi_enc.encode()` but the class exposes `embed_texts()`. This crashes the entire inference pipeline.

Fix: global search and replace

```
# Find every occurrence:
grep -rn 'poi_enc.encode\|poi_encoder.encode' app/ scripts/

# Replace with:
poi_enc.embed_texts(texts) # correct method name
```

Time estimate: 15 minutes

P0-4 Add Validation Monitoring During Training

File: scripts/train_multimodal.py

train_test_split creates a val set (line 54) but it is never used. The training loop has no val_loss, so overfitting is undetectable.

Fix: add val evaluation loop

```
# After the training loop, add:
model.eval()
with torch.no_grad():
    val_preds = model(val_X)
    val_loss = criterion(val_preds, val_y)
    val_acc = (val_preds.argmax(1) == val_y).float().mean()

# Also save training history for defense presentation:
history = {'train_loss': train_losses, 'val_loss': val_losses}
import json
json.dump(history, open('evals/training_history.json', 'w'))
```

Time estimate: 30 minutes

P0-5 Pre-Download OSM Road Network (Prevent Demo Crash)

File: app/infrastructure/road_network.py

No roads.graphml is cached. Without it, the pipeline downloads from OSM at demo time — a 10-30 second delay that can fail on conference WiFi.

Fix: pre-download before the defense

```
python - <<'EOF'
import osmnx as ox
G = ox.graph_from_bbox(30.10, 29.90, 31.30, 31.10, network_type='drive')
ox.save_graphml(G, 'data/raw/roads.graphml')
print('Done - road network cached')
EOF
```

Run this once at home on a good internet connection, commit the file, and the demo will load instantly.

Time estimate: 10 minutes (mostly download time)

3. P1 — Recommended Fixes (Should Complete Before Defense)

These improve your evaluation score from 22 to approximately 30-35/50 and demonstrate academic rigor.

P1-6 Add Ablation Study Support (--modalities flag)

Files: scripts/train_multimodal.py, app/application/fusion_service.py

The Step 2 student document explicitly requires ablation studies (POI only / Image only / All). Without a modality flag, this cannot be demonstrated.

Fix: add argparse CLI flag

```
import argparse
parser = argparse.ArgumentParser()
parser.add_argument('--modalities', default='poi,image,graph',
                    help='Comma-separated: poi,image,graph,text')
args = parser.parse_args()
modalities = args.modalities.split(',')

# Then conditionally build feature vector:
features = []
if 'poi' in modalities: features.append(poi_emb)
if 'image' in modalities: features.append(img_emb)
if 'graph' in modalities: features.append(graph_feat)
X = np.concatenate(features, axis=1)
```

Run three experiments: poi-only, poi+image, all. Save results to separate CSVs. Show the comparison table in your defense slides.

Time estimate: 1 hour

P1-7 Add Missing Output Schema Fields

File: app/application/fusion_service.py, around line 133-145

PRD v3.0 requires graph_embedding_norm and text_embedding_norm in each cell result. Currently missing (~78% schema compliance).

Fix: add two lines to the result dict

```
result = {
    'cell_id': cell_id,
    'dominant_class': dominant_class,
    'confidences': confidences,
    'poi_top_categories': poi_top_cats,
    'road_density_km_per_km2': road_density,
    'node_count': node_count,
    'satellite_thumbnail_url': thumb_url,
    'graph_embedding_norm': float(np.linalg.norm(graph_feat)), # ADD
    'text_embedding_norm': float(np.linalg.norm(poi_emb)),      # ADD
}
```

Time estimate: 30 minutes

P1-8 Save Training History to JSON

File: scripts/train_multimodal.py

No loss curve data is persisted. You cannot show a training convergence chart in your defense without this.

Fix: accumulate and save loss per epoch

```
train_losses = []
for epoch in range(epochs):
    # ... training loop ...
    train_losses.append(float(loss.item()))

import json, os
os.makedirs('evals', exist_ok=True)
json.dump({'train_loss': train_losses, 'val_loss': val_losses},
          open('evals/training_history.json', 'w'))
```

Time estimate: 15 minutes

P1-9 Implement Spatial Accuracy Metric

File: evals/eval_multimodal.py

Required by PRD v3.0 FR-29. Spatial Accuracy measures the % of cells where the predicted class matches the majority class among its 8 geographic neighbors — a key urban planning metric.

Fix: implement 8-neighbor voting

```
import numpy as np

def spatial_accuracy(pred_grid: np.ndarray) -> float:
    """
    pred_grid: 2D array of predicted class labels (shape: H x W)
    Returns: % cells consistent with their 8 neighbors
    """
    H, W = pred_grid.shape
    correct = 0
    for r in range(H):
        for c in range(W):
            neighbors = []
            for dr in [-1, 0, 1]:
                for dc in [-1, 0, 1]:
                    if dr==0 and dc==0: continue
                    nr, nc = r+dr, c+dc
                    if 0<=nr<H and 0<=nc<W:
                        neighbors.append(pred_grid[nr, nc])
            if neighbors:
                majority = max(set(neighbors), key=neighbors.count)
                if pred_grid[r,c] == majority: correct += 1
    return correct / (H * W)
```

Time estimate: 2 hours

P1-10 Add DELETE Job Endpoint

File: app/interfaces/api.py

Required by PRD v3.0 FR-20. Needed by the frontend Cancel button during classification.

Fix: add one endpoint

```
@router.delete('/api/v1/jobs/{job_id}')
async def cancel_job(job_id: str):
```

```
if job_id in running_jobs:
    running_jobs[job_id]['status'] = 'cancelled'
    return {'status': 'cancelled', 'job_id': job_id}
raise HTTPException(status_code=404, detail='Job not found')
```

Time estimate: 30 minutes

4. P2 — Future Work (Mention in Defense as Planned Enhancements)

These are architecturally complex and beyond what is expected for an MVP defense. Frame them as 'Version 2 roadmap' in your presentation.

Feature	Description	Reference
GNN (PyTorch Geometric)	Replace MLP with GraphSAGE or GATConv for spatial reasoning over road graph	PRD FR-13, Phase 7
Full MLLM Integration	LLaMA-3.2-1B or DistilGPT-2 with LoRA fine-tuning for geo-language tasks	PRD FR-34, Phase 8
Digital Twin NL Query	Arabic/English query → GeoJSON cell highlights with spatial explanation	PRD FR-46, Phase 11
Attention Fusion	Replace concat with cross-modal attention to learn modality weights dynamically	PRD FR-12, Phase 6
Geo-MLLM Export	Export trained model as .pt / .gguf / .onnx with auto-generated model card	PRD FR-44, Phase 12
Knowledge Distillation	Compress large LLM into small geo-aware model for edge deployment	Student Doc Phase 12

5. Step 2 Phase Completion Map

Cross-reference of your 12-phase student plan against the audit findings.

#	Phase	Status	Remaining Tasks
2	Multi-Modal Data Collection	PARTIAL	Pre-cache roads.graphml (P0-5). No other blockers.
3	Data Cleaning (WGS84)	OK	Backend only — no UI needed. Done.
4	Graph $G=(V,E)$	PARTIAL	OSMnx integration exists. Cache GraphML (P0-5). GET /graph-topology endpoint missing.
6	Feature Fusion (643 dim)	OK	Concat fusion works. Add ablation flag (P1-6). Add attention fusion as P2.
7	Base Models (ResNet + MiniLM + MLP)	PARTIAL	ResNet18 + MiniLM working. MLP weight mismatch (P0-2). GNN is P2.
9	Training Artifacts	PARTIAL	Fix encode→embed_texts (P0-3). Add val monitoring (P0-4). Save history JSON (P1-8).
10	Evaluation Metrics	PARTIAL	Fix dataset labels first (P0-1). Then: add Spatial Accuracy (P1-9). Add 2 schema fields (P1-7).
11	Digital Twin NL Query	MISSING	Returns 501. Full implementation is P2 — mention in defense as planned.
12	Geo-MLLM Export + Demo	MISSING	P2 — no .pt/.gguf/.onnx export. No model card. Mention as roadmap.

6. Prioritized Fix Timeline

Priority	ID	File	Task	Hours	Dependency
P0	1	data/raw/project.csv	Re-label CSV with multi-class ground truth	2.0	None
P0	2	domain/mlp_model.py: 12	Fix hidden_dim=256 or retrain	0.5	After P0-1
P0	3	scripts/train_multimodal.py	encode() → embed_texts()	0.25	None
P0	4	scripts/train_multimodal.py	Add val_loss monitoring loop	0.5	None
P0	5	data/raw/roads.graphml	Pre-download and cache OSM road network	0.1	None
P1	6	train_multimodal.py	Add --modalities argparse flag	1.0	None
P1	7	fusion_service.py	Add graph/text embedding norm to output schema	0.5	None
P1	8	train_multimodal.py	Save training history to JSON	0.25	P0-4
P1	9	evals/eval_multimodal.py	Implement 8-neighbor Spatial Accuracy metric	2.0	P0-1
P1	10	interfaces/api.py	Add DELETE /api/v1/jobs/{job_id}	0.5	None
P2	11 +	Multiple	GNN / MLLM / Digital Twin / LoRA / Export	20+	Post-defense

P0 Total (Must-Fix)	P1 Total (Should-Fix)	Combined Total
~3.35 hours	~4.25 hours	~7.6 hours

7. API Coverage Gaps

9 of 16 PRD v3.0 endpoints are implemented (56%). The table below shows what is missing.

Endpoint	PRD Ref	Status	Action
GET /grid/{id}/graph-topology	FR-08	MISSING	Needed for GNN layer on map. Medium priority.
DELETE /api/v1/jobs/{job_id}	FR-20	MISSING	Simple — add cancel endpoint (P1-10)
POST /api/v1/evaluate	FR-28	MISSING	Needed for Spatial Accuracy metric (P1-9)
GET /api/v1/evaluate/{id}/export	FR-62	MISSING	Add after /evaluate is working
POST /api/v1/mllm/train	FR-40	MISSING	P2 — full MLLM builder, post-defense
GET /api/v1/mllm/status/{id}	FR-41	MISSING	P2 — dependent on mllm/train
GET /api/v1/mllm/export/{id}	FR-44	MISSING	P2 — dependent on mllm/train
POST /api/v1/query (NL query)	FR-48	MISSING	Returns 501. P2 — Digital Twin, post-defense.

8. Projected Score After Fixes

Dimension	Current	After P0	After P0+P1
ML Training Quality	3/10	7/10	8/10
Output Schema	7/10	7/10	9/10
API Coverage	6/10	6/10	7/10
Phase Completion	5/10	6/10	8/10
Demo Stability	1/10	7/10	8/10
TOTAL	22/50	33/50	40/50

P0 alone brings you from 22 to ~33 — just below the 35-point threshold but defensible with strong presentation of limitations.

P0 + P1 brings you to ~40 — above the threshold and a solid academic defense.